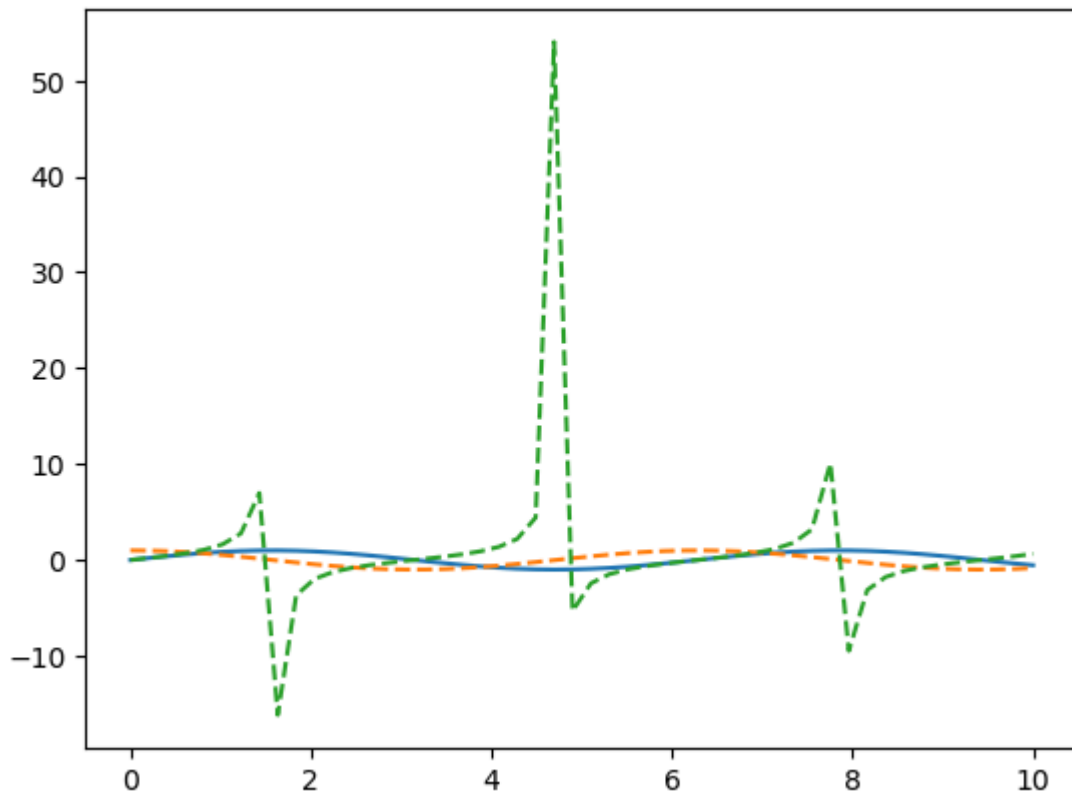


```
In [6]: import numpy as np
import matplotlib.pyplot as plt
```

```
In [13]: x1=np.linspace(0,10,50)
plt.plot(x1,np.sin(x1),'-')
plt.plot(x1,np.cos(x1),'--')
plt.plot(x1,np.tan(x1),'--')
plt.show()
```



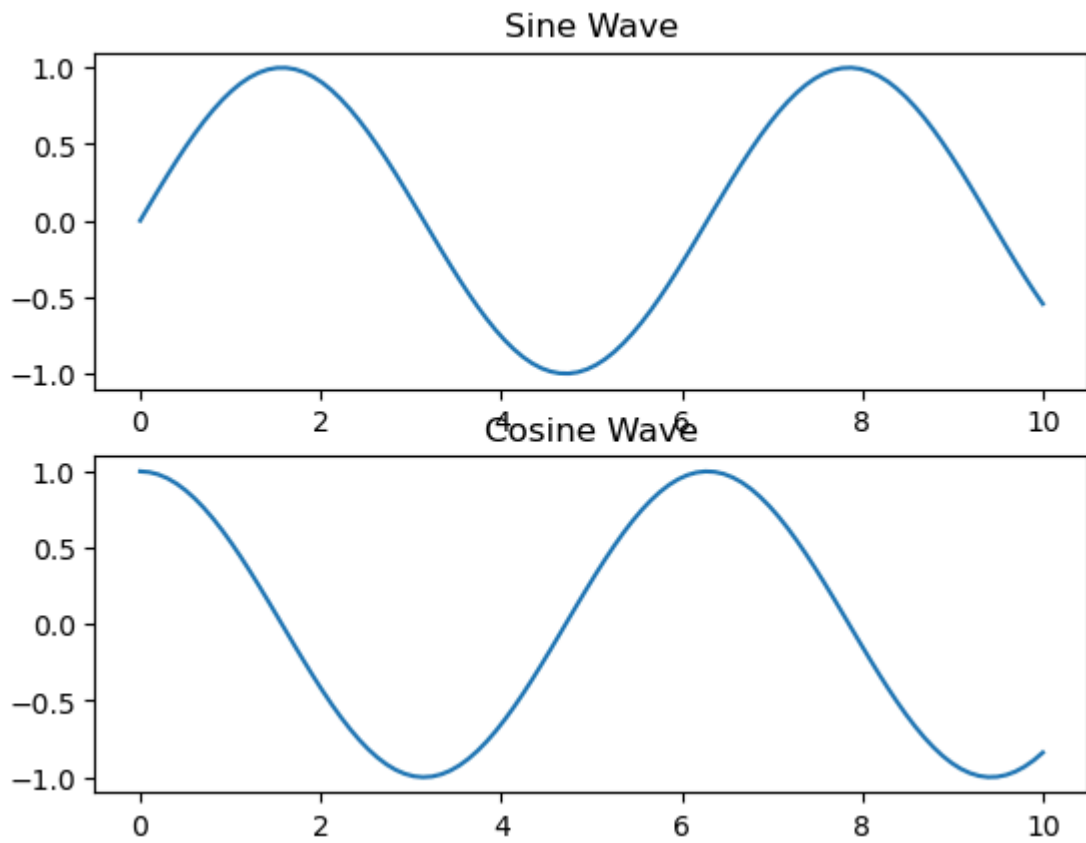
```
In [18]: import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, 10, 100)

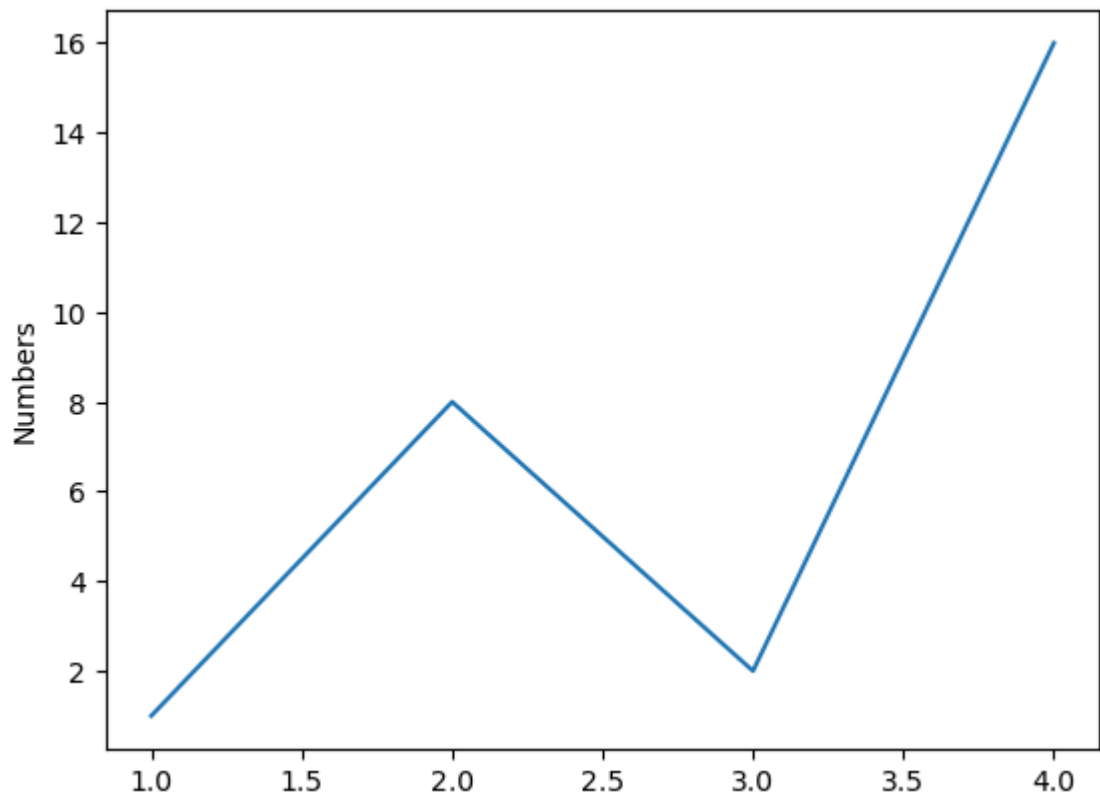
# First subplot
plt.subplot(2, 1, 1)
plt.plot(x, np.sin(x))
plt.title("Sine Wave")

# Second subplot
plt.subplot(2, 1, 2)
plt.plot(x, np.cos(x))
plt.title("Cosine Wave")

plt.show()
```

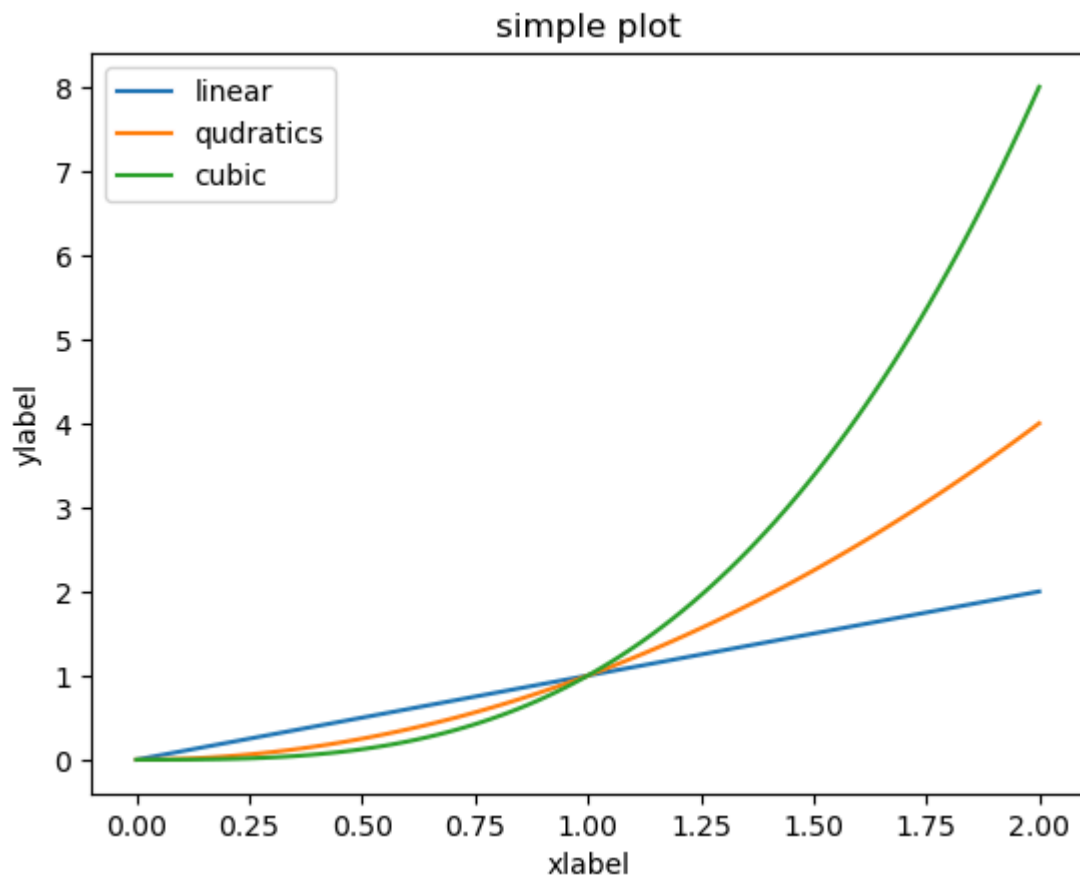


```
In [21]: plt.plot([1,2,3,4],[1,8,2,16])  
plt.ylabel('Numbers')  
plt.show()
```



```
In [5]: import numpy as np  
import matplotlib.pyplot as plt  
x=np.linspace(0,2,100)  
plt.plot(x,x,label='linear')
```

```
plt.plot(x,x**2,label='qudratics')
plt.plot(x,x**3,label='cubic')
plt.xlabel('xlabel')
plt.ylabel('ylabel')
plt.title('simple plot')
plt.legend('simple plot')
plt.legend()
plt.show()
```



```
In [7]: plt.plot([1,2,3,4],[1,4,9,16], 'ro')
plt.axis([0,6,0,20])
plt.show()
```

ValueError

Traceback (most recent call last)

Cell In[7], line 1

```
----> 1 plt.plot([1,2,3,4],[1,4,916], )
      2 plt.axis([0,6,0,20])
      3 plt.show()
```

File ~\anaconda3\Lib\site-packages\matplotlib\pyplot.py:3838, in plot(scalex, scaley, data, *args, **kwargs)

```
3830 @_copy_docstring_and_deprecators(Axes.plot)
3831 def plot(
3832     *args: float | ArrayLike | str,
3833     (...) 3836     **kwargs,
3837 ) -> list[Line2D]:
-> 3838     return gca().plot(
3839         *args,
3840         scalex=scalex,
3841         scaley=scaley,
3842         **({ : data} if data is not None else {}),
3843         **kwargs,
3844     )
```

File ~\anaconda3\Lib\site-packages\matplotlib\axes_axes.py:1777, in Axes.plot(self, scalex, scaley, data, *args, **kwargs)

```
1534 """
1535 Plot y versus x as lines and/or markers.
1536
1537 (...) 1774 (``'green'``) or hex strings (``'#008000'``).
1775 """
1776 kwargs = cbook.normalize_kwargs(kwargs, mlines.Line2D)
-> 1777 lines = [*self._get_lines(self, *args, data=data, **kwargs)]
1778 for line in lines:
1779     self.add_line(line)
```

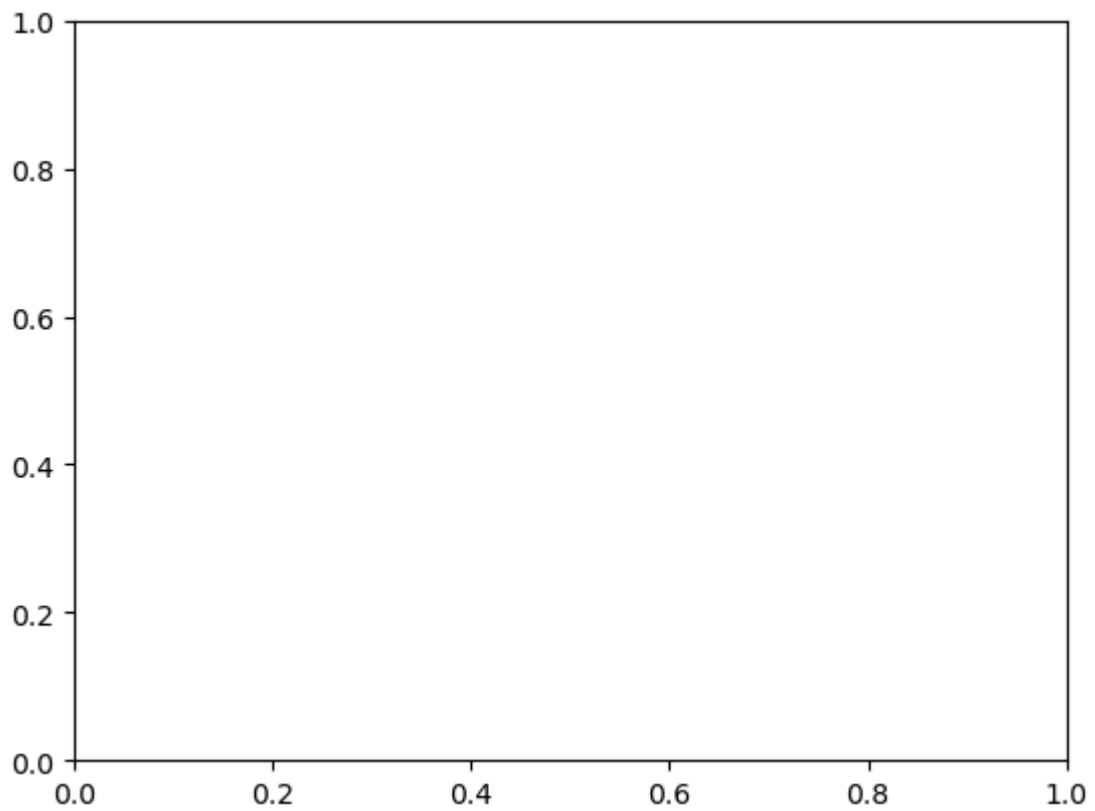
File ~\anaconda3\Lib\site-packages\matplotlib\axes_base.py:297, in _process_plot_var_args.__call__(self, axes, data, return_kwargs, *args, **kwargs)

```
295     this += args[0],
296     args = args[1:]
--> 297 yield from self._plot_args(
298     axes, this, kwargs, ambiguous_fmt_datakey=ambiguous_fmt_datakey,
299     return_kwargs=return_kwargs
300 )
```

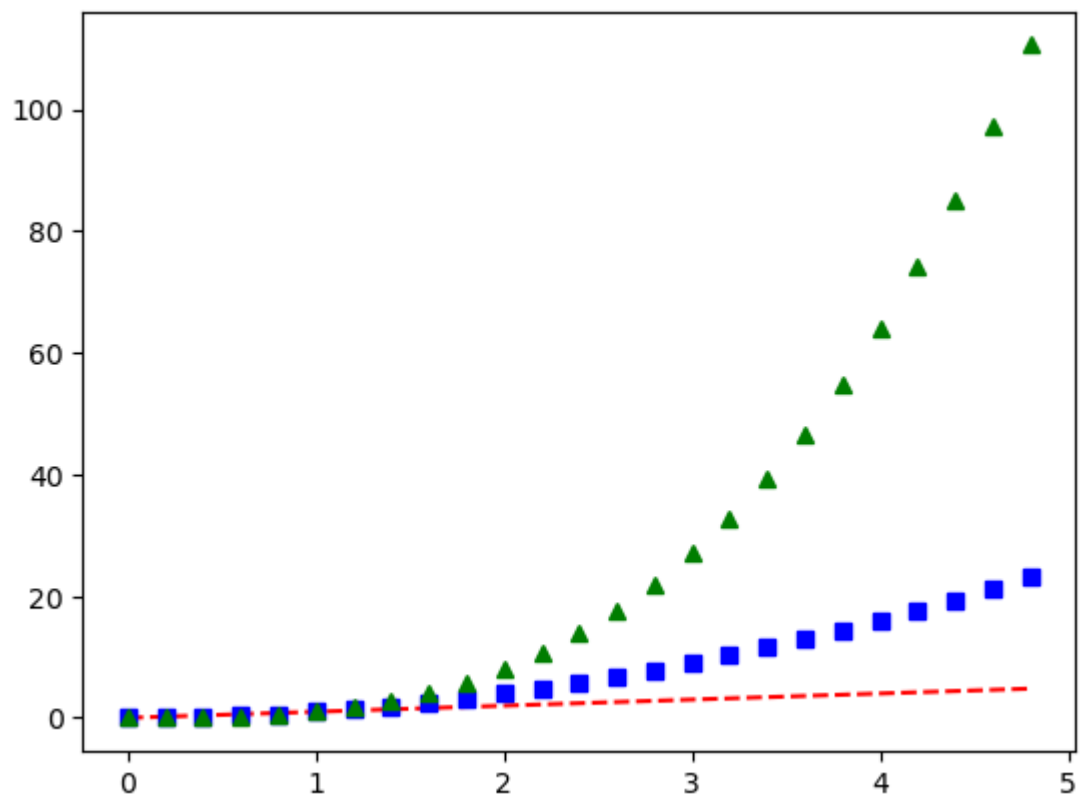
File ~\anaconda3\Lib\site-packages\matplotlib\axes_base.py:494, in _process_plot_var_args._plot_args(self, axes, tup, kwargs, return_kwargs, ambiguous_fmt_datakey)

```
491     axes.yaxis.update_units(y)
493 if x.shape[0] != y.shape[0]:
--> 494     raise ValueError(f"x and y must have same first dimension, but "
495                     f"have shapes {x.shape} and {y.shape}")
496 if x.ndim > 2 or y.ndim > 2:
497     raise ValueError(f"x and y can be no greater than 2D, but have "
498                     f"shapes {x.shape} and {y.shape}")
```

ValueError: x and y must have same first dimension, but have shapes (4,) and (3,)



```
In [8]: t=np.arange(0.,5.0,0.2)
plt.plot(t,t,'r--',t,t**2,'bs',t,t**3,'g^')
plt.show()
```



```
In [13]: fig=plt.figure()
x2=np.linspace(0,5,10)
y2=x**2
axes=fig.add_axes([0.1,0.1,0.8,0.8])
axes.plot(x2,y2,'r')
```

```
axes.set_xlabel('x2')
axes.set_ylabel('y2')
axes.set_title('title')
```

ValueError Traceback (most recent call last)

Cell In[13], line 5

```
3 y2=x**2
4 axes=fig.add_axes([0.1,0.1,0.8,0.8])
----> 5 axes.plot(x2,y2, )
6 axes.set_xlabel('x2')
7 axes.set_ylabel('y2')
```

File ~\anaconda3\Lib\site-packages\matplotlib\axes_axes.py:1777, in Axes.plot(self, scalex, scaley, data, *args, **kwargs)

```
1534 """
1535 Plot y versus x as lines and/or markers.
1536 (...) 1774 (``'green'``) or hex strings (``'#008000'``).
1775 """
1776 kwargs = cbook.normalize_kwargs(kwargs, mlines.Line2D)
-> 1777 lines = [*self._get_lines(self, *args, data=data, **kwargs)]
1778 for line in lines:
1779     self.add_line(line)
```

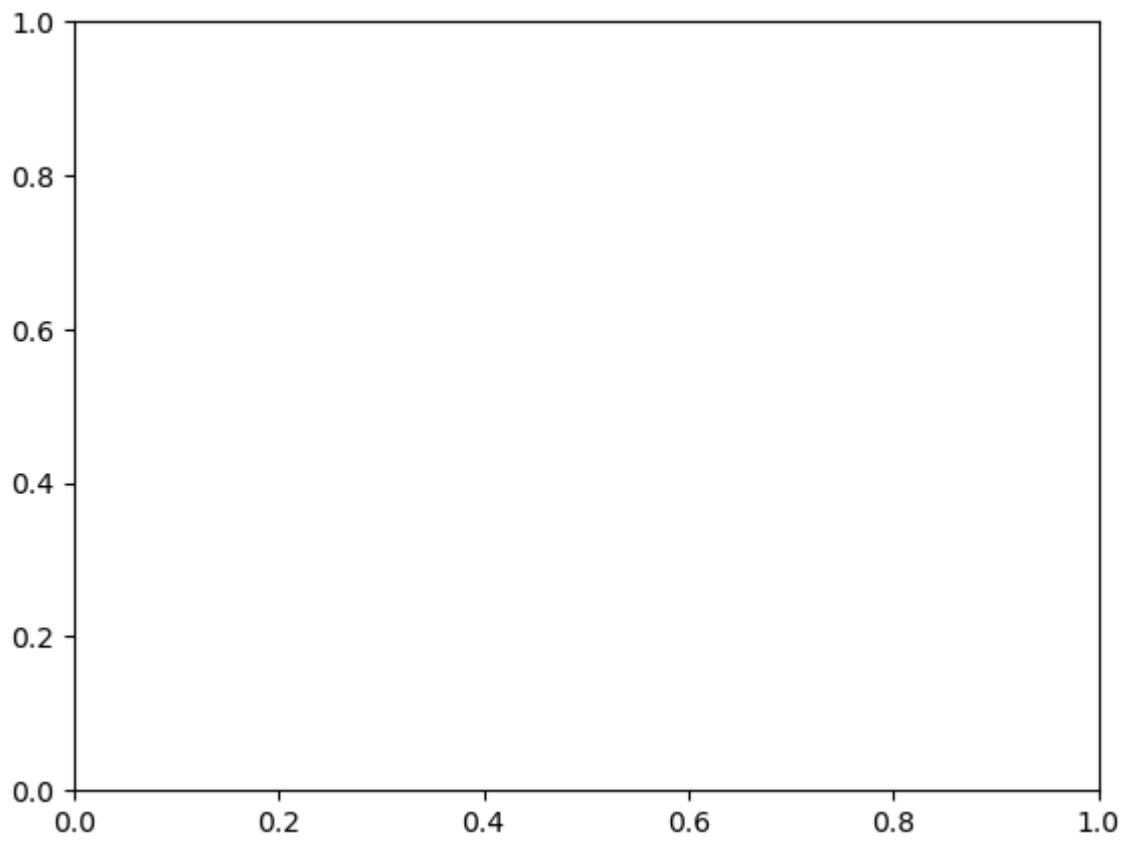
File ~\anaconda3\Lib\site-packages\matplotlib\axes_base.py:297, in _process_plot_var_args.__call__(self, axes, data, return_kwargs, *args, **kwargs)

```
295     this += args[0],
296     args = args[1:]
--> 297 yield from self._plot_args(
298     axes, this, kwargs, ambiguous_fmt_datakey=ambiguous_fmt_datakey,
299     return_kwargs=return_kwargs
300 )
```

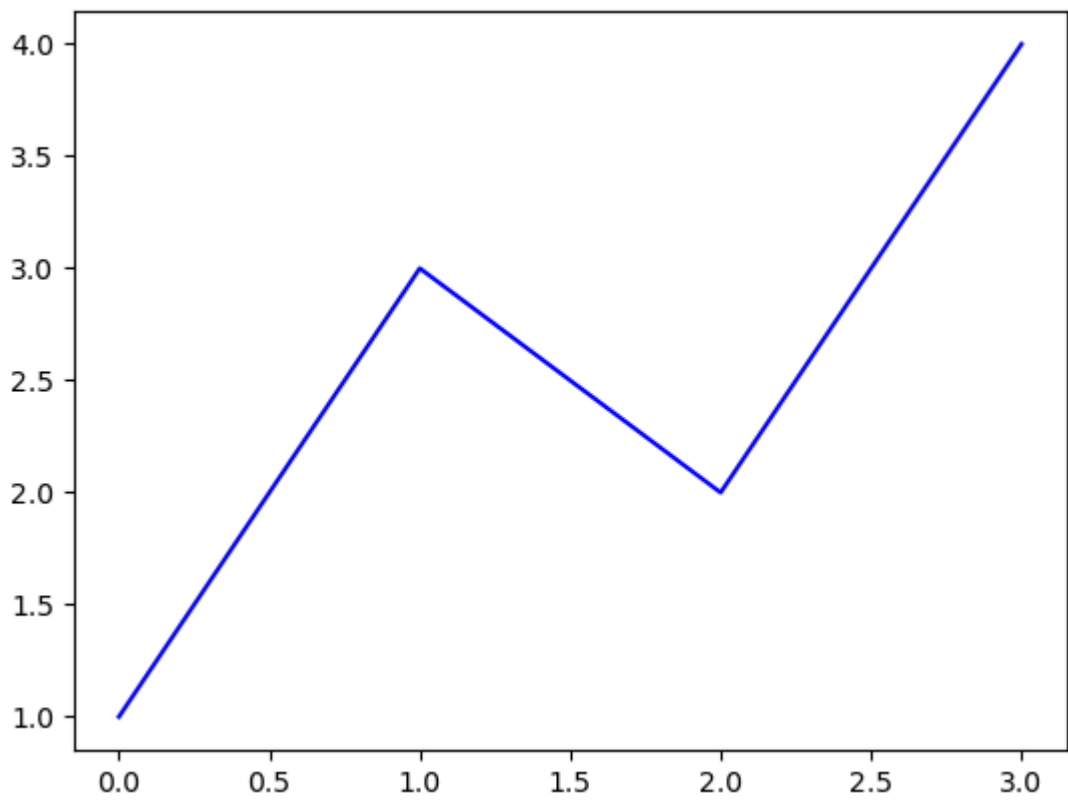
File ~\anaconda3\Lib\site-packages\matplotlib\axes_base.py:494, in _process_plot_var_args._plot_args(self, axes, tup, kwargs, return_kwargs, ambiguous_fmt_datakey)

```
491     axes.yaxis.update_units(y)
493 if x.shape[0] != y.shape[0]:
--> 494     raise ValueError(f"x and y must have same first dimension, but "
495                     f"have shapes {x.shape} and {y.shape}")
496 if x.ndim > 2 or y.ndim > 2:
497     raise ValueError(f"x and y can be no greater than 2D, but have "
498                     f"shapes {x.shape} and {y.shape}")
```

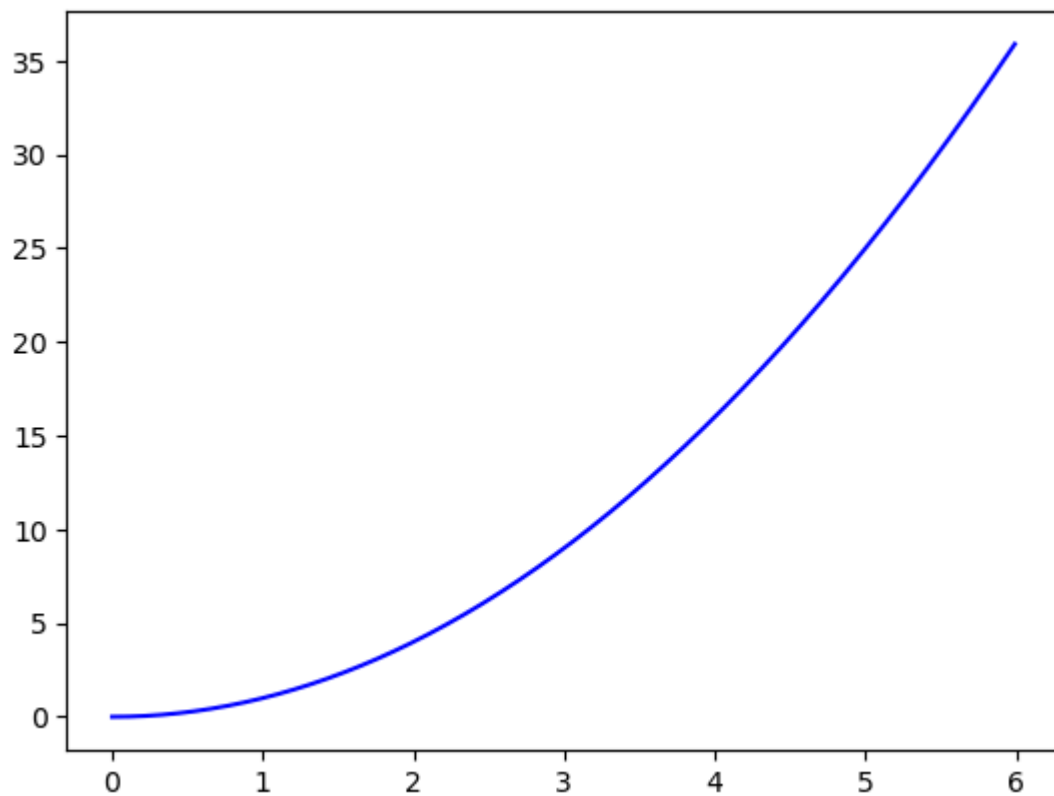
ValueError: x and y must have same first dimension, but have shapes (10,) and (10,)



```
In [19]: plt.plot([1,3,2,4], 'b-')  
plt.show()
```

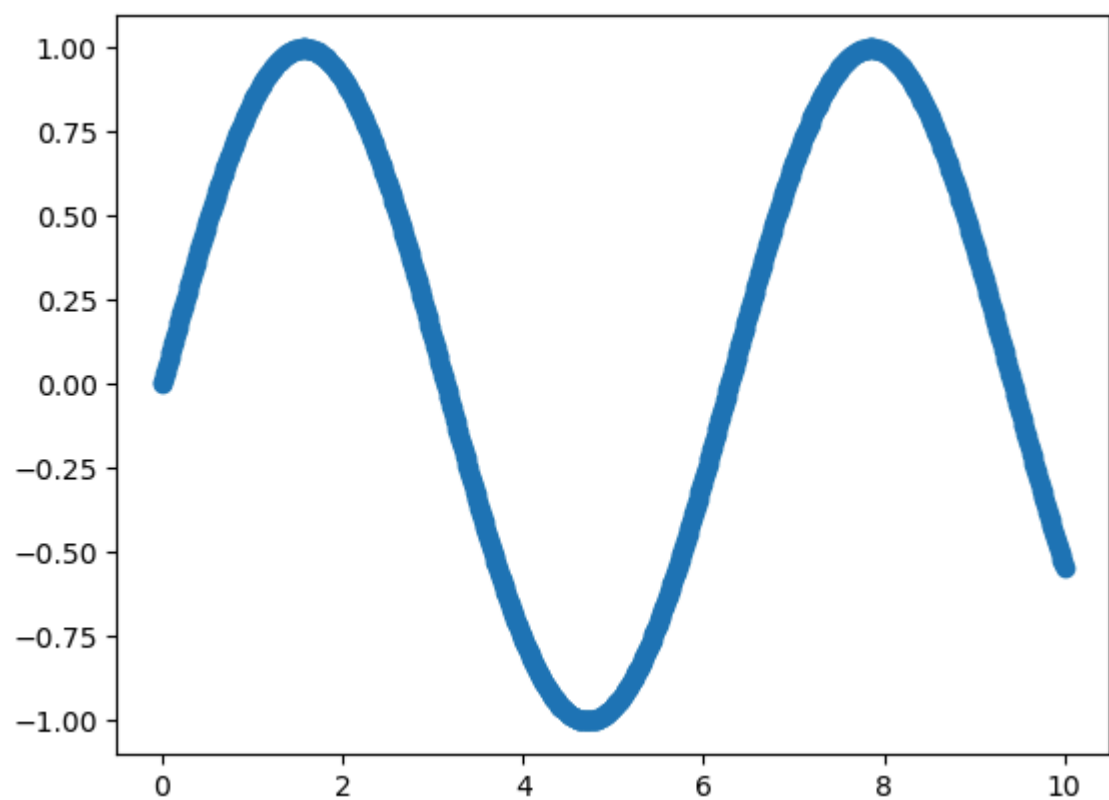


```
In [21]: x3=np.arange(0.0,6.0,0.01)  
plt.plot(x3,[xi**2 for xi in x3], 'b-')  
plt.show()
```



```
In [26]: fig=plt.figure()
ax=plt.axes()
x5=np.linspace(0,10,1000)
ax.plot(x5,np.sin(x5),'o')
```

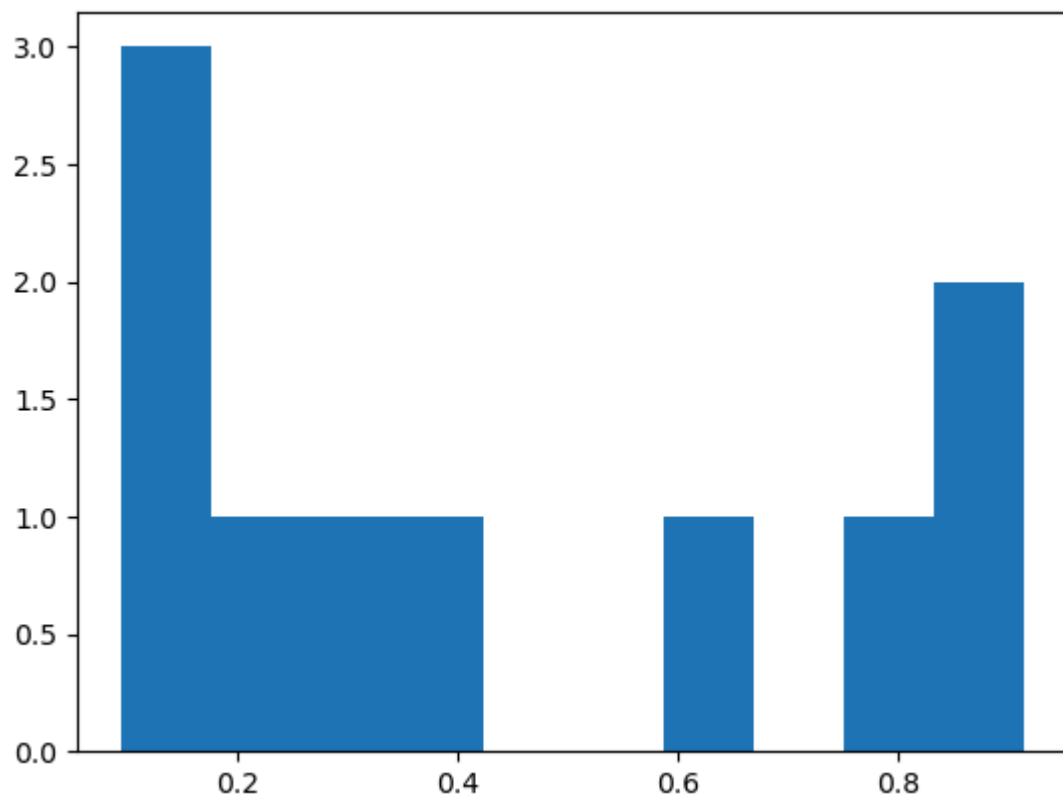
Out[26]: [



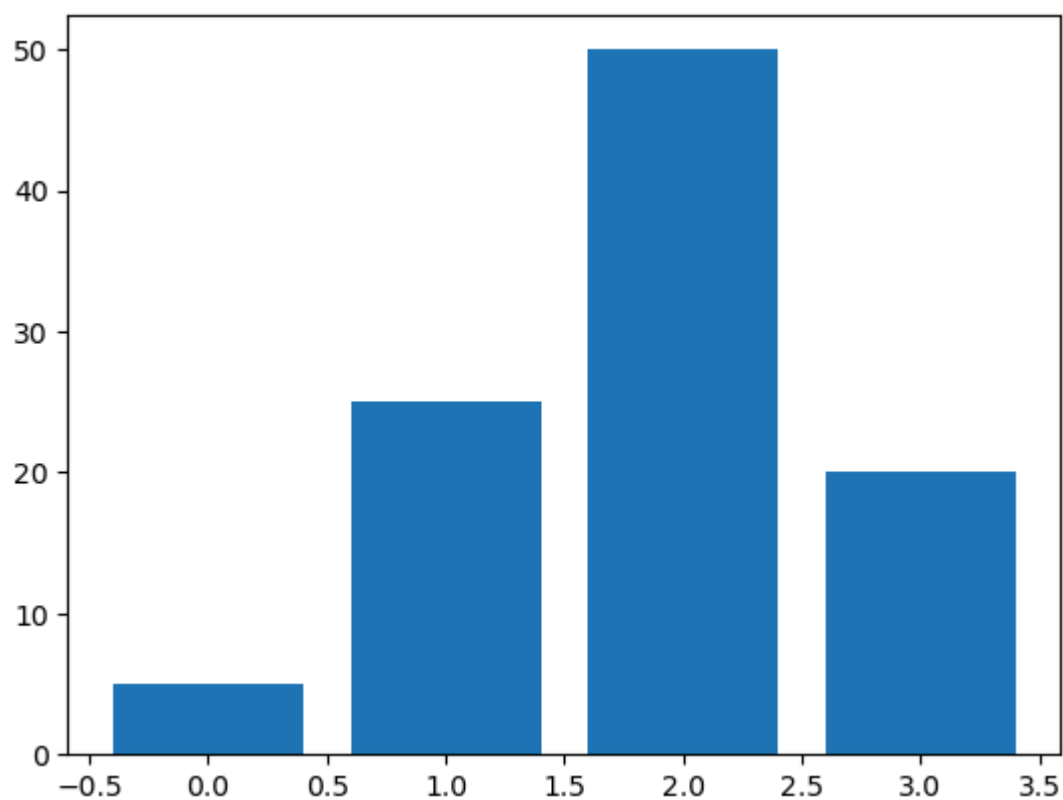
```
In [29]: data1=np.random.rand(10)
plt.hist(data1)
```



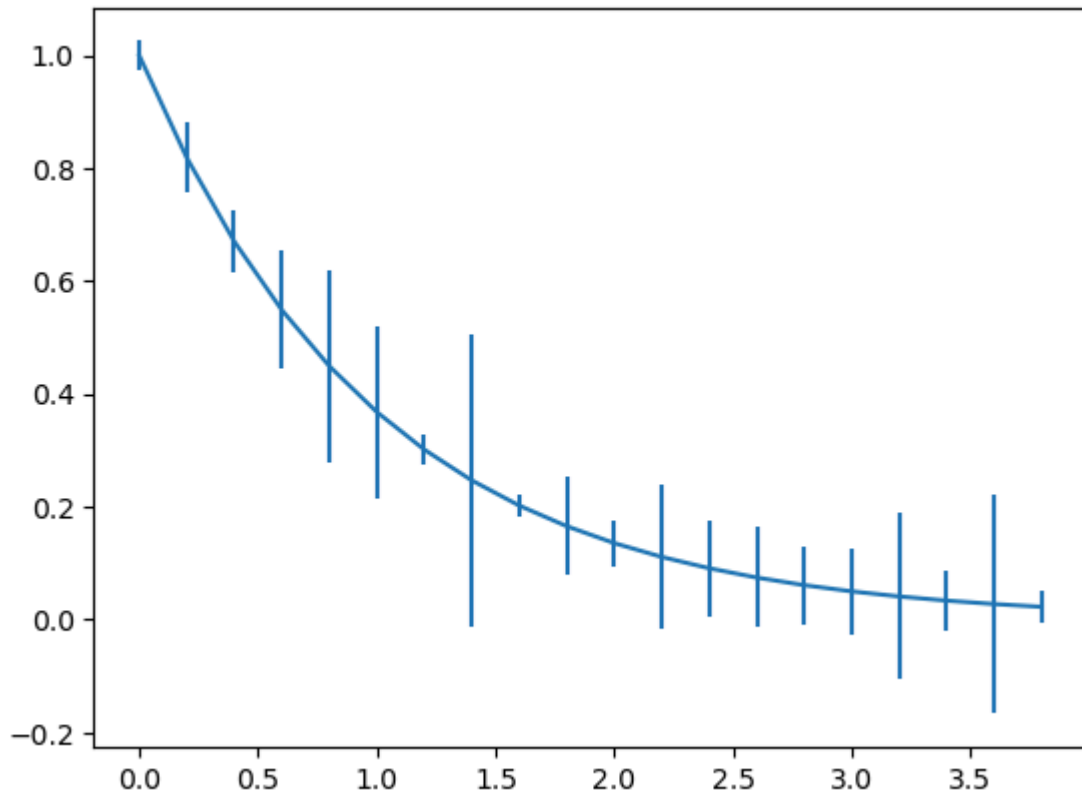
```
Out[29]: (array([3., 1., 1., 1., 0., 0., 1., 0., 1., 2.]),
          array([0.09547336, 0.17748941, 0.25950546, 0.3415215 , 0.42353755,
                0.50555359, 0.58756964, 0.66958568, 0.75160173, 0.83361778,
                0.91563382])),
          <BarContainer object of 10 artists>)
```



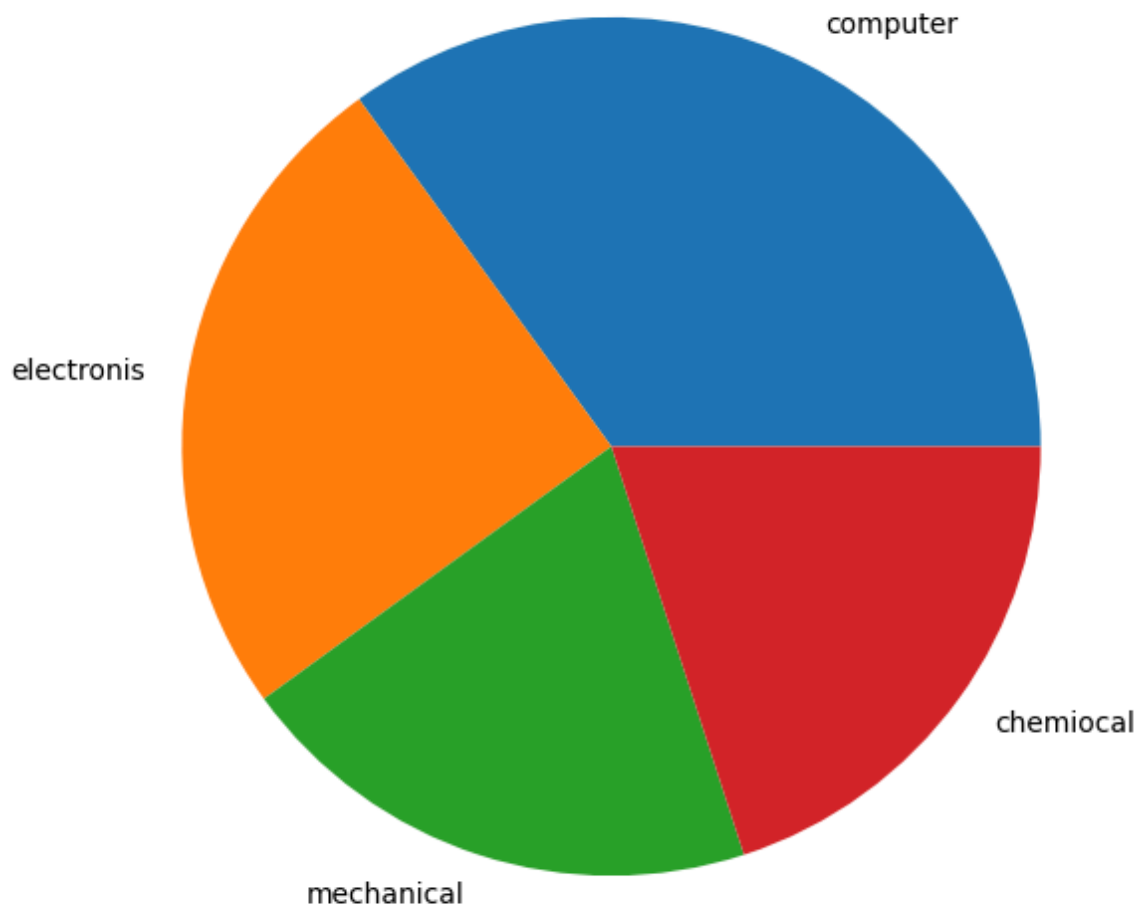
```
In [30]: data2=[5.,25.,50.,20.,]
plt.bar(range(len(data2)),data2)
plt.show()
```



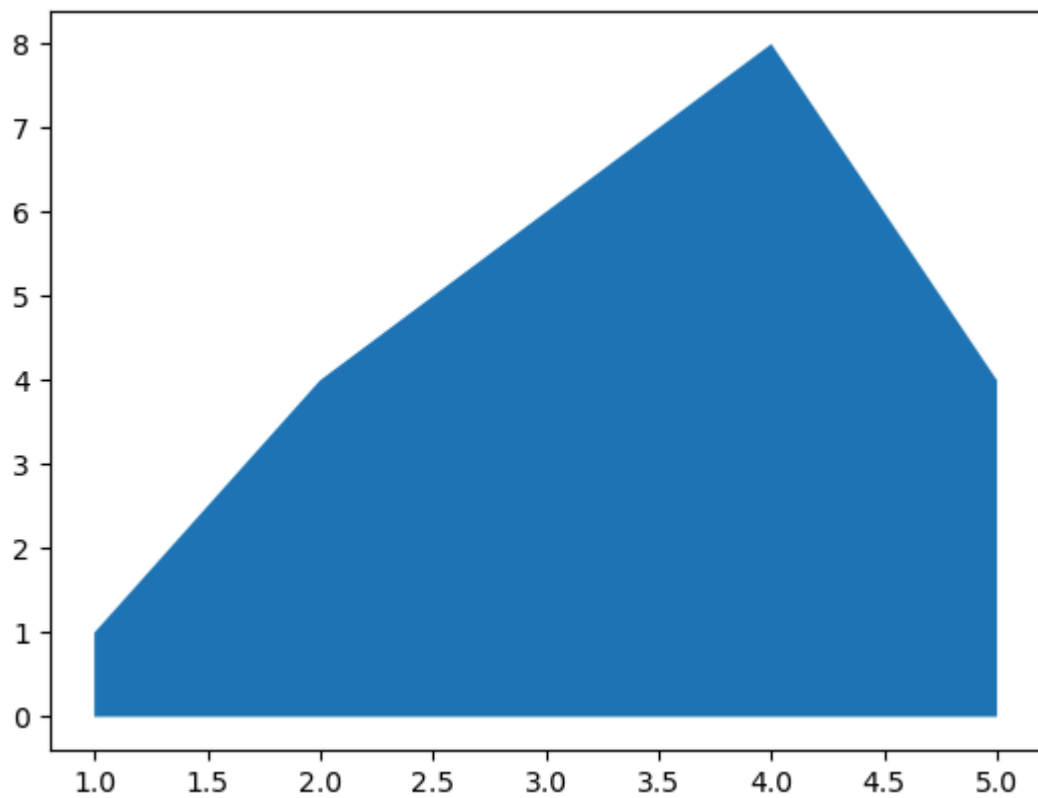
```
In [31]: x9=np.arange(0,4,0.2)
y9=np.exp(-x9)
e1=0.1*np.abs(np.random.randn(len(y9)))
plt.errorbar(x9,y9,yerr=e1,fmt='-')
plt.show()
```



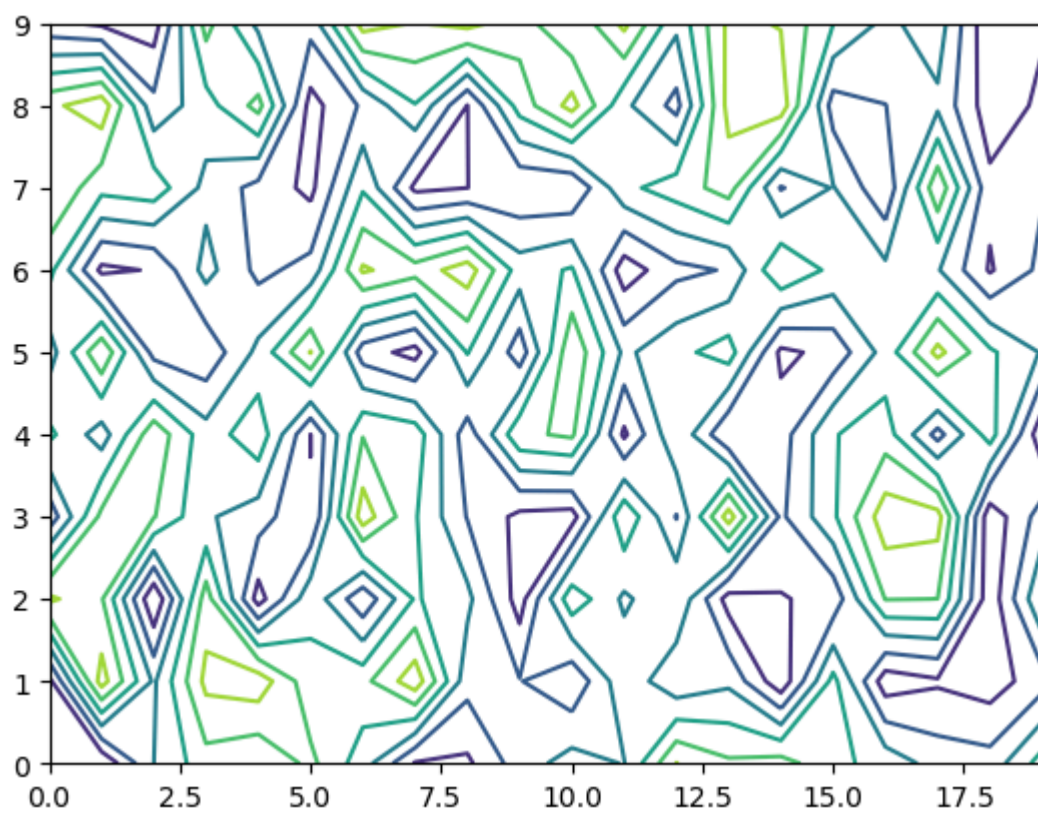
```
In [32]: plt.figure(figsize=(7,7))
x10=[35,25,20,20]
labels=['computer','electronic','mechanical','chemical']
plt.pie(x10,labels=labels)
plt.show()
```



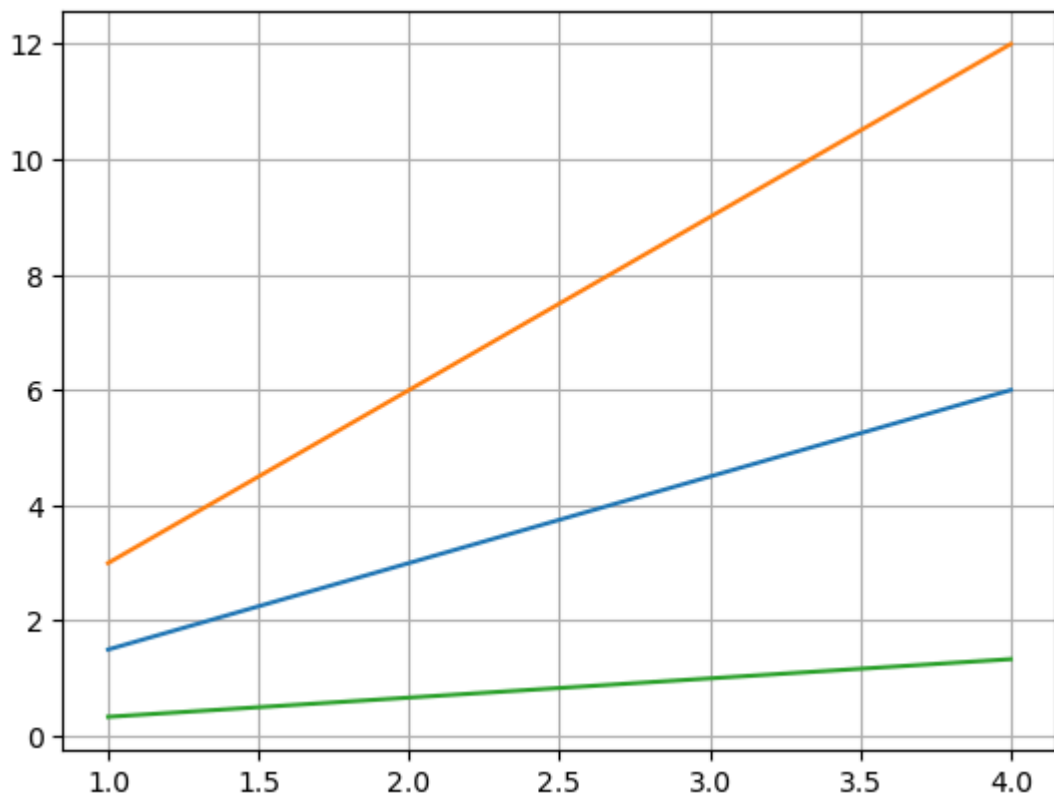
```
In [33]: x12=range(1,6)
          y12=[1,4,6,8,4]
          plt.fill_between(x12,y12)
          plt.show()
```



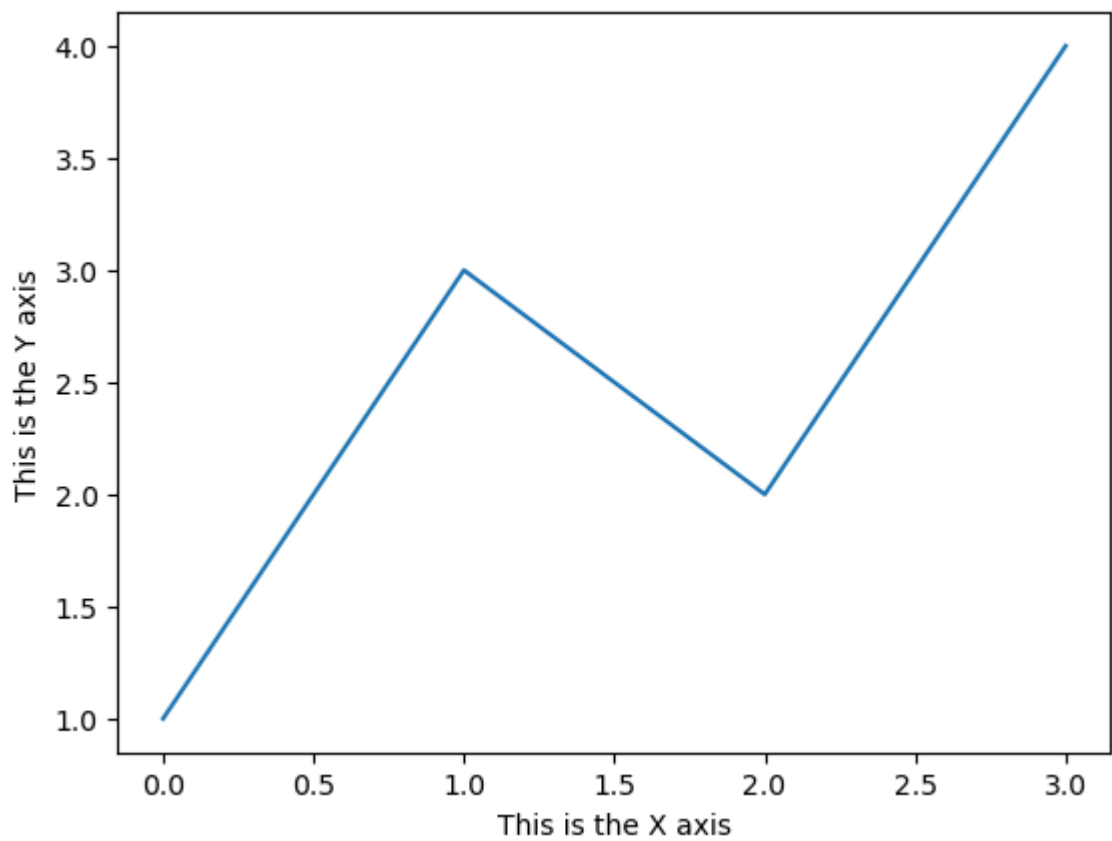
```
In [34]: matrix1=np.random.rand(10,20)
cp=plt.contour(matrix1)
plt.show()
```



```
In [35]: x15=np.arange(1,5)
plt.plot(x15,x15*1.5,x15,x15*3.0,x15,x15/3.0)
plt.grid(True)
plt.show()
```



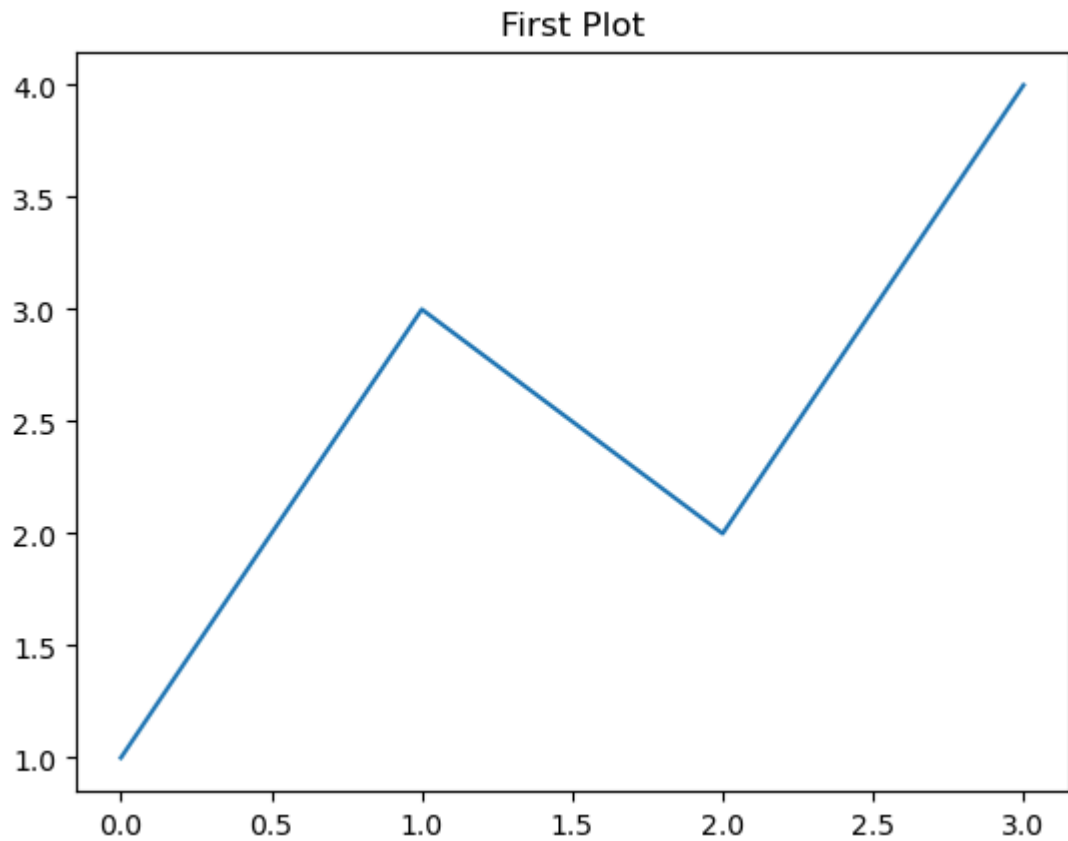
```
In [36]: plt.plot([1, 3, 2, 4])  
  
plt.xlabel('This is the X axis')  
  
plt.ylabel('This is the Y axis')  
  
plt.show()
```



```
In [37]: plt.plot([1, 3, 2, 4])

plt.title('First Plot')

plt.show()
```

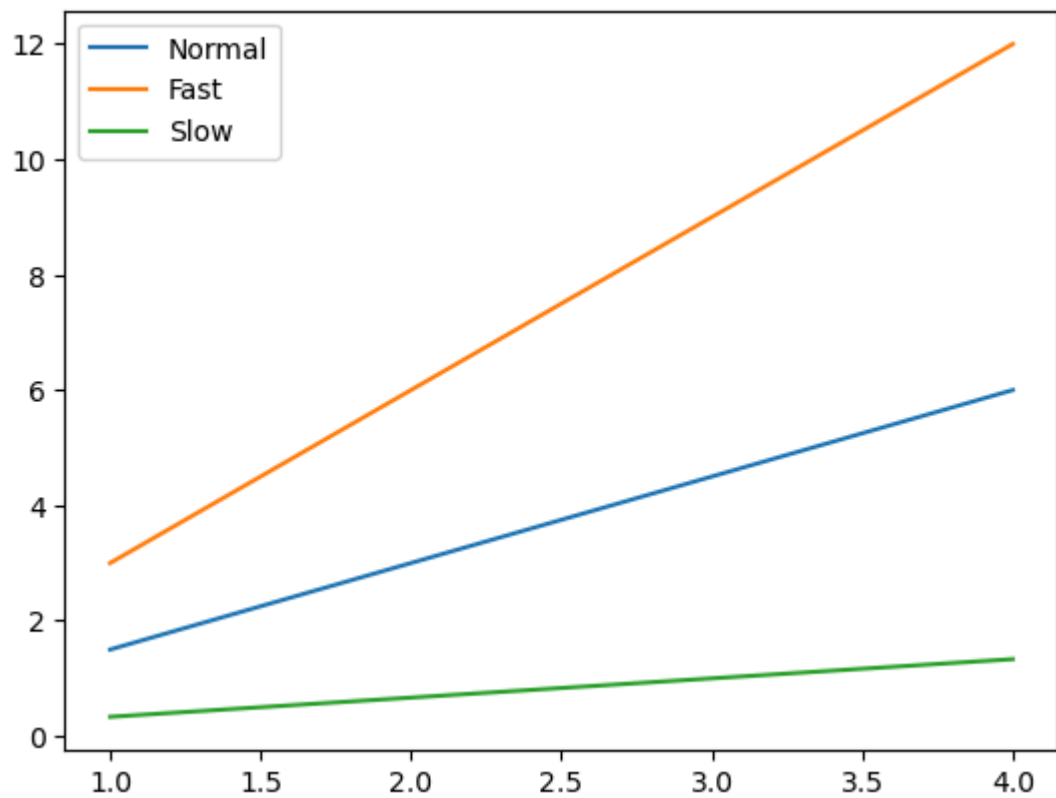


```
In [38]: x15 = np.arange(1, 5)

fig, ax = plt.subplots()

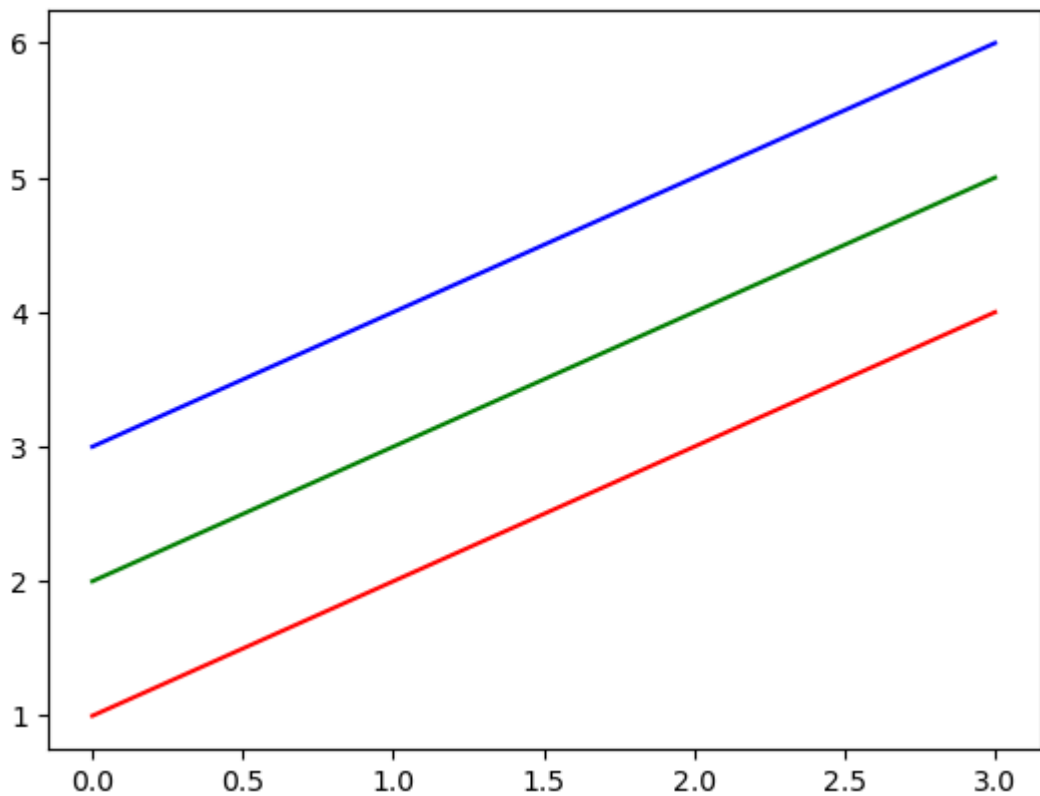
ax.plot(x15, x15*1.5)
ax.plot(x15, x15*3.0)
ax.plot(x15, x15/3.0)

ax.legend(['Normal', 'Fast', 'Slow']);
```



```
In [39]: x16 = np.arange(1, 5)
```

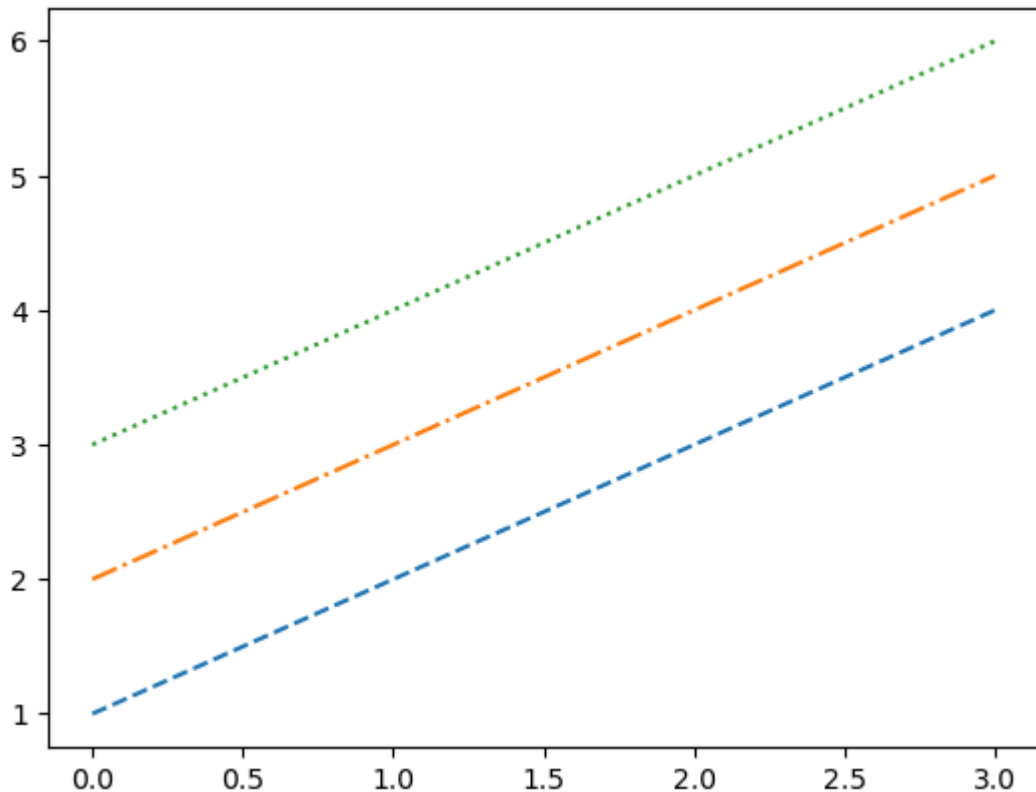
```
plt.plot(x16, 'r')  
plt.plot(x16+1, 'g')  
plt.plot(x16+2, 'b')  
  
plt.show()
```



```
In [40]: x16 = np.arange(1, 5)

plt.plot(x16, '--', x16+1, '-.', x16+2, ':')

plt.show()
```



```
In [ ]: #datasets matplotlib
```

```
In [41]: #Import numpy
import numpy as np

#Seasons
Seasons = ["2015", "2016", "2017", "2018", "2019", "2020", "2021", "2022", "2023", "2024"]
Sdict = {"2015":0, "2016":1, "2017":2, "2018":3, "2019":4, "2020":5, "2021":6, "2022":7

#Players
Players = ["Sachin", "Rahul", "Smith", "Sami", "Pollard", "Morris", "Samson", "Dhoni", "
Pdct = {"Sachin":0, "Rahul":1, "Smith":2, "Sami":3, "Pollard":4, "Morris":5, "Samson":

#Salaries
Sachin_Salary = [15946875, 17718750, 19490625, 21262500, 23034375, 24806250, 25244493,
Rahul_Salary = [12000000, 12744189, 13488377, 14232567, 14976754, 16324500, 18038573, 1
Smith_Salary = [4621800, 5828090, 13041250, 14410581, 15779912, 14500000, 16022500, 175
Sami_Salary = [3713640, 4694041, 13041250, 14410581, 15779912, 17149243, 18518574, 1945
Pollard_Salary = [4493160, 4806720, 6061274, 13758000, 15202590, 16647180, 18091770, 19
Morris_Salary = [3348000, 4235220, 12455000, 14410581, 15779912, 14500000, 16022500, 17
Samson_Salary = [3144240, 3380160, 3615960, 4574189, 13520500, 14940153, 16359805, 1777
Dhoni_Salary = [0, 0, 4171200, 4484040, 4796880, 6053663, 15506632, 16669630, 17832627, 1
Kohli_Salary = [0, 0, 0, 4822800, 5184480, 5546160, 6993708, 16402500, 17632688, 18862875
Sky_Salary = [3031920, 3841443, 13041250, 14410581, 15779912, 14200000, 15691000, 17182

#Matrix
Salary = np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary, Polla

#Games
```



```

Sachin_G = [80,77,82,82,73,82,58,78,6,35]
Rahul_G = [82,57,82,79,76,72,60,72,79,80]
Smith_G = [79,78,75,81,76,79,62,76,77,69]
Sami_G = [80,65,77,66,69,77,55,67,77,40]
Pollard_G = [82,82,82,79,82,78,54,76,71,41]
Morris_G = [70,69,67,77,70,77,57,74,79,44]
Samson_G = [78,64,80,78,45,80,60,70,62,82]
Dhoni_G = [35,35,80,74,82,78,66,81,81,27]
Kohli_G = [40,40,40,81,78,81,39,0,10,51]
Sky_G = [75,51,51,79,77,76,49,69,54,62]
#Matrix
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G, Samson_G, Dhoni_G, Kohli_G, Sky_G])

#Points
Sachin_PTS = [2832,2430,2323,2201,1970,2078,1616,2133,83,782]
Rahul_PTS = [1653,1426,1779,1688,1619,1312,1129,1170,1245,1154]
Smith_PTS = [2478,2132,2250,2304,2258,2111,1683,2036,2089,1743]
Sami_PTS = [2122,1881,1978,1504,1943,1970,1245,1920,2112,966]
Pollard_PTS = [1292,1443,1695,1624,1503,1784,1113,1296,1297,646]
Morris_PTS = [1572,1561,1496,1746,1678,1438,1025,1232,1281,928]
Samson_PTS = [1258,1104,1684,1781,841,1268,1189,1186,1185,1564]
Dhoni_PTS = [903,903,1624,1871,2472,2161,1850,2280,2593,686]
Kohli_PTS = [597,597,597,1361,1619,2026,852,0,159,904]
Sky_PTS = [2040,1397,1254,2386,2045,1941,1082,1463,1028,1331]
#Matrix
Points = np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS, Morris_PTS, Samson_PTS, Dhoni_PTS, Kohli_PTS, Sky_PTS])

```

In [43]: Salary

```

Out[43]: array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
                25244493, 27849149, 30453805, 23500000],
                [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
                18038573, 19752645, 21466718, 23180790],
                [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
                [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
                18518574, 19450000, 22407474, 22458000],
                [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
                18091770, 19536360, 20513178, 21436271],
                [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
                [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
                16359805, 17779458, 18668431, 20068563],
                [      0,      0,  4171200,  4484040,  4796880,  6053663,
                15506632, 16669630, 17832627, 18995624],
                [      0,      0,      0,  4822800,  5184480,  5546160,
                6993708, 16402500, 17632688, 18862875],
                [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
                15691000, 17182000, 18673000, 15000000]])

```

In [44]: Games

```
Out[44]: array([[80, 77, 82, 82, 73, 82, 58, 78, 6, 35],
               [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
               [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
               [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
               [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
               [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
               [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
               [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
               [40, 40, 40, 81, 78, 81, 39, 0, 10, 51],
               [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

```
In [45]: Sdict
```

```
Out[45]: {'2015': 0,
          '2016': 1,
          '2017': 2,
          '2018': 3,
          '2019': 4,
          '2020': 5,
          '2021': 6,
          '2022': 7,
          '2023': 8,
          '2024': 9}
```

```
In [46]: Pdict
```

```
Out[46]: {'Sachin': 0,
          'Rahul': 1,
          'Smith': 2,
          'Sami': 3,
          'Pollard': 4,
          'Morris': 5,
          'Samson': 6,
          'Dhoni': 7,
          'Kohli': 8,
          'Sky': 9}
```

```
In [47]: Salary//Games
```

```
C:\Users\honey\AppData\Local\Temp\ipykernel_19252\1634212085.py:1: RuntimeWarning:
divide by zero encountered in floor_divide
Salary//Games
```

```
Out[47]: array([[ 199335,  230113,  237690,  259298,  315539,  302515,  435249,
                  357040,  5075634,  671428],
                [ 146341,  223582,  164492,  180159,  197062,  226729,  300642,
                  274342,  271730,  289759],
                [  58503,   74719,  173883,  177908,  207630,  183544,  258427,
                  230855,  247629,  299194],
                [  46420,   72216,  169366,  218342,  228694,  222717,  336701,
                  290298,  291006,  561450],
                [  54794,   58618,   73917,  174151,  185397,  213425,  335032,
                  257057,  288918,  522835],
                [  47828,   61380,  185895,  187150,  225427,  188311,  281096,
                  237094,  241360,  469190],
                [  40310,   52815,   45199,   58643,  300455,  186751,  272663,
                  253992,  301103,  244738],
                [     0,     0,   52140,   60595,   58498,   77611,  234948,
                  205797,  220155,  703541],
                [     0,     0,     0,   59540,   66467,   68471,  179325,
                  0, 1763268,  369860],
                [  40425,   75322,  255710,  182412,  204933,  186842,  320224,
                  249014,  345796,  241935]])
```

```
In [48]: Games[:5]
```

```
Out[48]: array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
                [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
                [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
                [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
                [82, 82, 82, 79, 82, 78, 54, 76, 71, 41]])
```

```
In [49]: Salary[0]
```

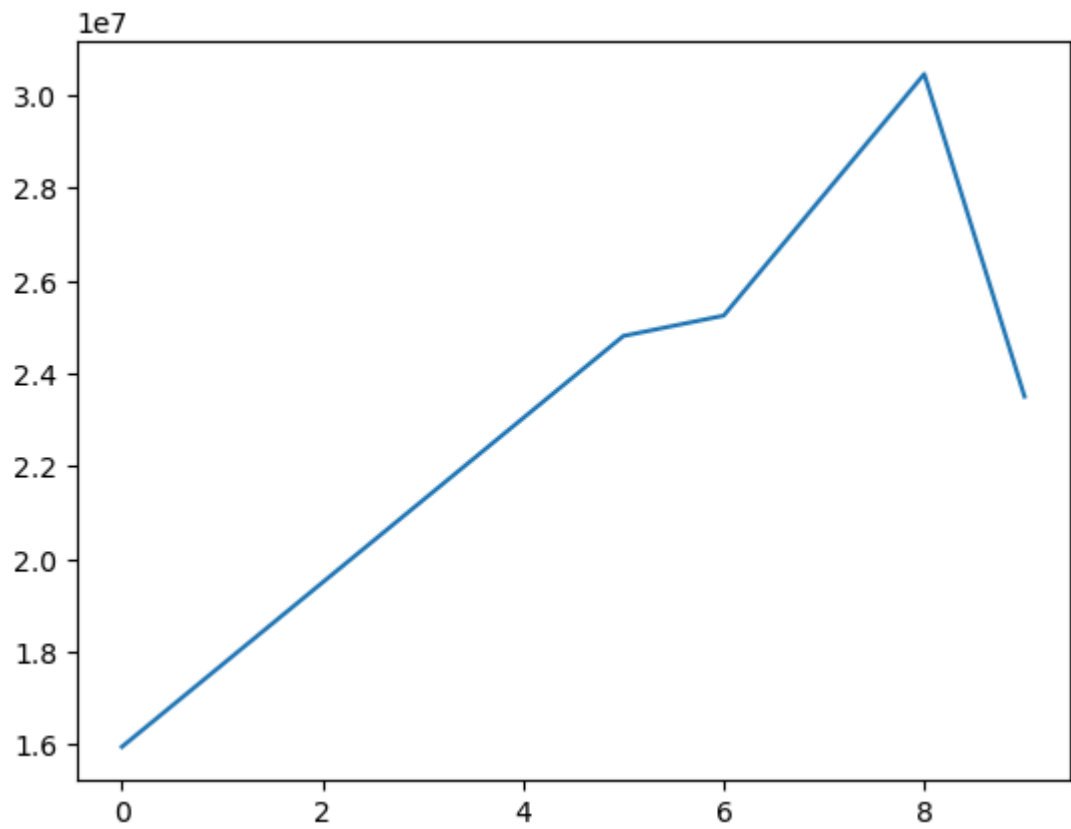
```
Out[49]: array([15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
                25244493, 27849149, 30453805, 23500000])
```

```
In [50]: import warnings
         warnings.filterwarnings('ignore')
```

```
In [51]: import matplotlib.pyplot as plt
```

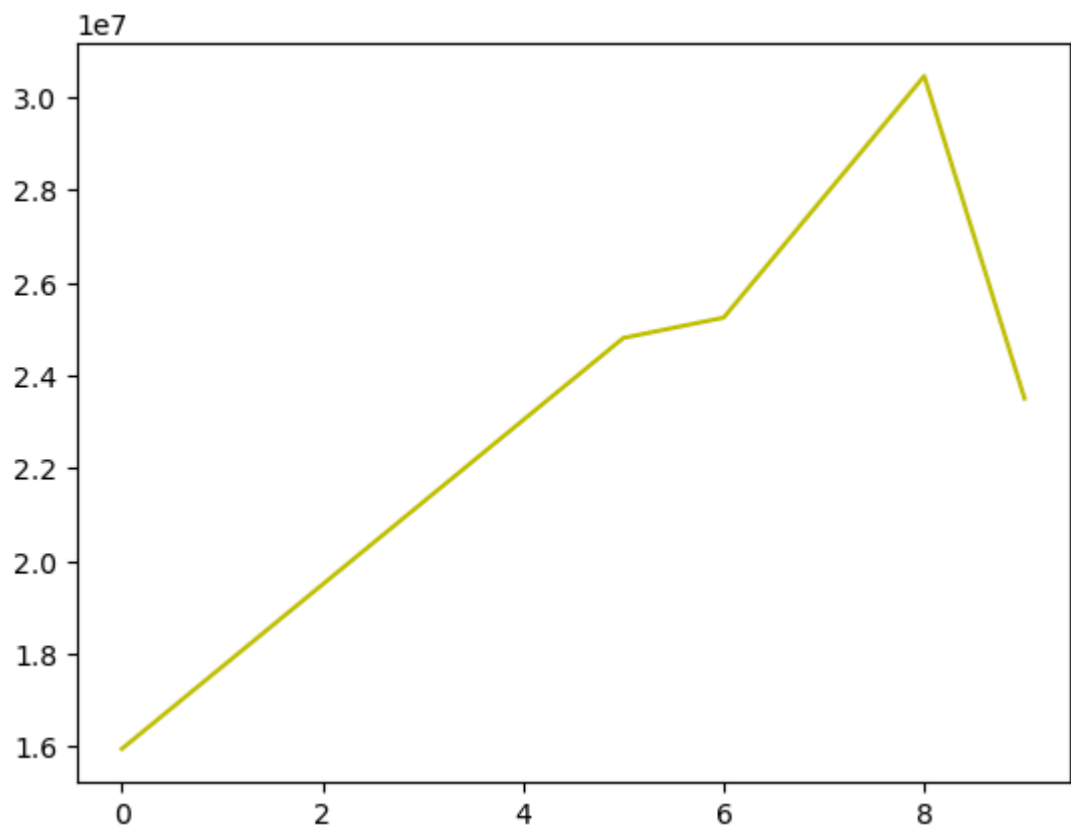
```
In [52]: plt.plot(Salary[0])
```

```
Out[52]: [<matplotlib.lines.Line2D at 0x27577491590>]
```



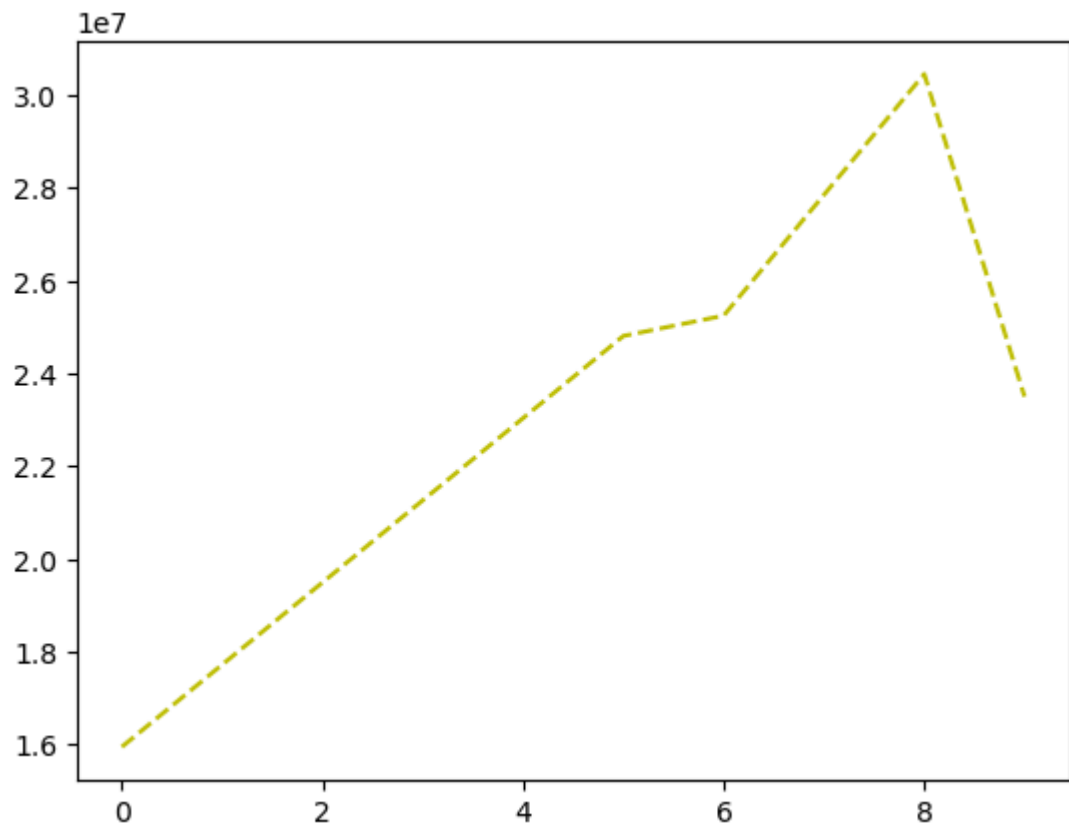
```
In [53]: plt.plot(Salary[0], c='y')
```

```
Out[53]: [<matplotlib.lines.Line2D at 0x2757713bd90>]
```



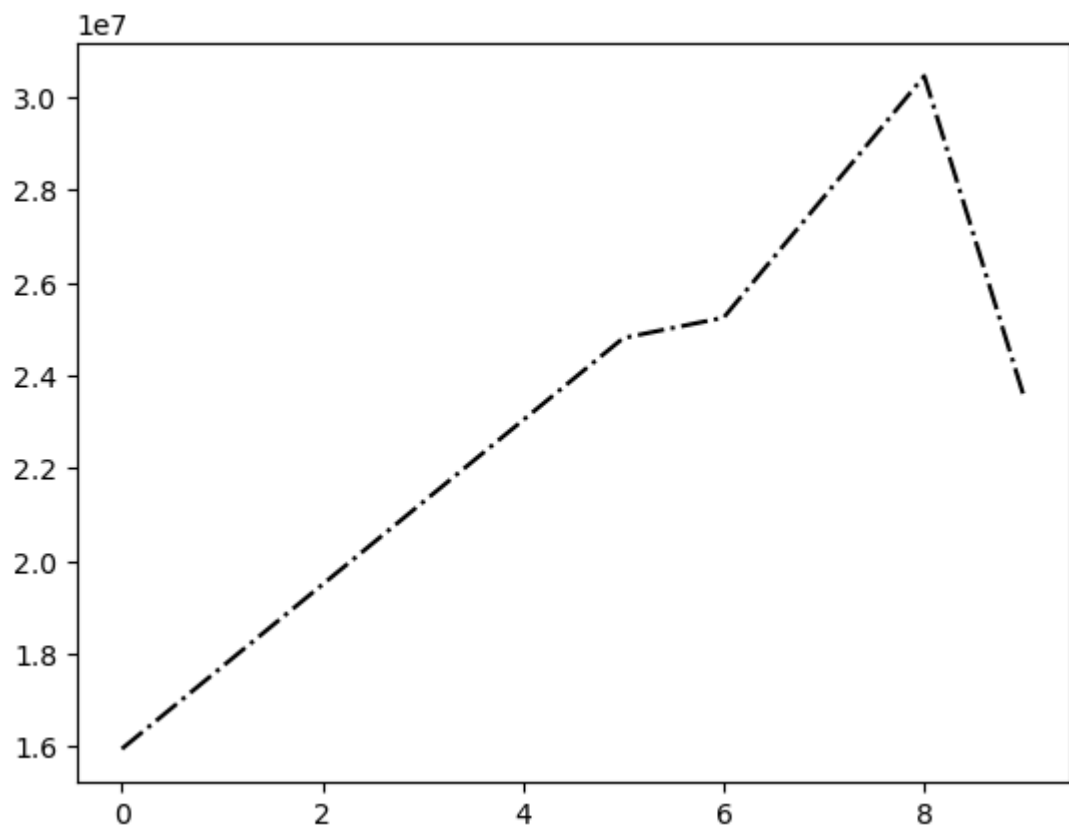
```
In [54]: plt.plot(Salary[0], c='y', ls = '--')
```

```
Out[54]: [<matplotlib.lines.Line2D at 0x27577294a50>]
```



```
In [55]: plt.plot(Salary[0], c='k', ls = '-.')
```

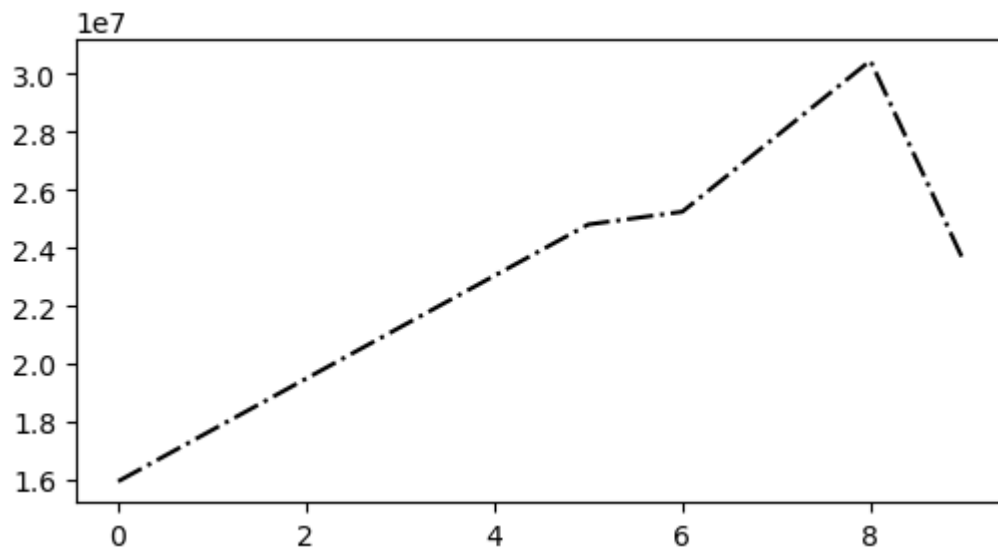
```
Out[55]: [<matplotlib.lines.Line2D at 0x27577114910>]
```



```
In [57]: %matplotlib inline
plt.rcParams['figure.figsize'] = 6,3
```

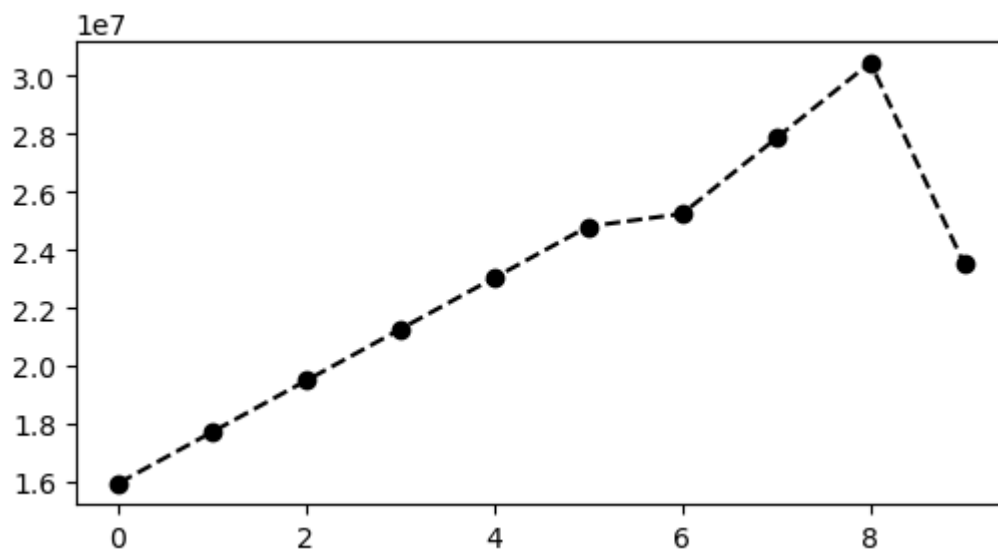
```
In [58]: plt.plot(Salary[0], c='k', ls = '-.')
```

```
Out[58]: [<matplotlib.lines.Line2D at 0x2757748ae90>]
```

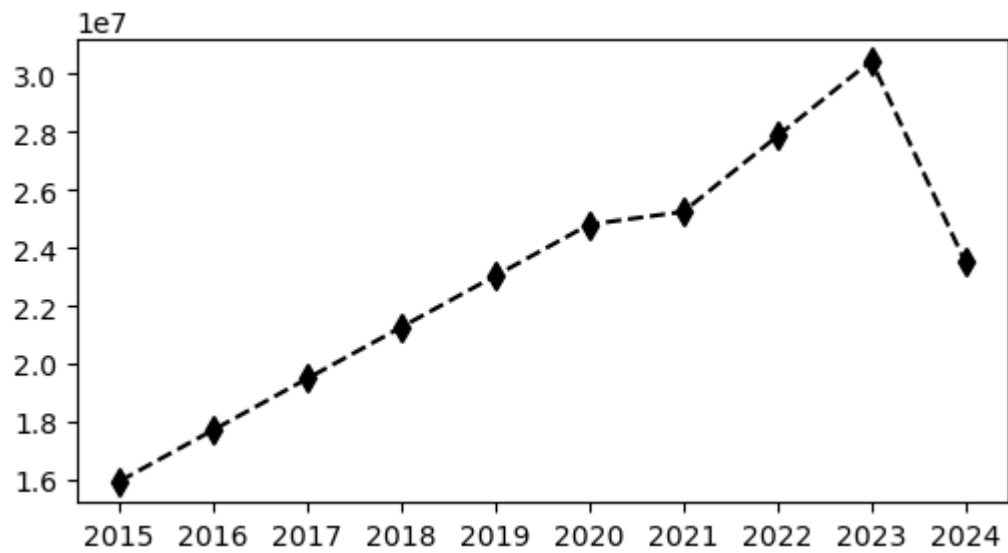


```
In [59]: plt.plot(Salary[0], c='k', ls = '--', marker = 'o')
```

```
Out[59]: [<matplotlib.lines.Line2D at 0x2757730d450>]
```



```
In [60]: plt.plot(Salary[0], c='black', ls = '--', marker = 'd', ms = 7)
plt.xticks(list(range(0,10)), Seasons)
plt.show()
```



```
In [61]: plt.plot(Salary[0], c='black', ls = '--', marker = 'd', ms = 7)
plt.xticks(list(range(0,10)), Seasons, rotation='vertical')
plt.show()
```

